

Austin Hoover

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Education and Employment

Accelerator Physicist, Oak Ridge National Laboratory	2024–
Postdoctoral Research Associate, Oak Ridge National Laboratory	2022–2024
Ph.D. in Physics, University of Tennessee	2018–2022
B.S. in Physics, Wheaton College (IL)	2014–2018

Journal Articles

<i>Demonstration of a novel phase space painting method in a coupled lattice to mitigate space charge in high-intensity hadron beams.</i> N. Evans, A. Hoover, T. Gorlov, and V. Morozov. Phys. Rev. Letters 136, 075002. [link] [arxiv] [data]	2026
<i>N-dimensional maximum-entropy tomography via particle sampling.</i> A. Hoover. Phys. Rev. Accel. Beams 28, L084601 (Letter, Editor's Choice). [link] [arxiv] [code]	2025
<i>Four-dimensional phase space tomography from one-dimensional measurements of a hadron beam.</i> A. Hoover. Phys. Rev. Accel. Beams 27, 122802. [link] [arxiv] [data]	2024
<i>High-dimensional maximum-entropy phase space tomography using normalizing flows.</i> A. Hoover and J.C. Wong. Phys. Rev. Research 6.3, 033163. [link] [arxiv] [code]	2024
<i>Analysis of a hadron beam in five-dimensional phase space.</i> A. Hoover, K. Ruisard, A. Aleksandrov, A. Zhukov, and S. Cousineau. Phys. Rev. Accel. Beams 26.6, 064202. [link] [arxiv] [data]	2023
<i>Four-dimensional emittance measurement at the Spallation Neutron Source.</i> A. Hoover and N. Evans. Nucl. Instrum. Methods Phys. Res., Sect. A 1041, 167376. [link] [arxiv]	2022
<i>Computation of the matched envelope of the Danilov distribution.</i> A. Hoover, N. Evans, and J. Holmes. Phys. Rev. Accel. Beams 24.4, 044201. [link]	2022

Conference/Workshop Proceedings

Invited Talks

<i>Eigenpainting for mitigation of space charge in high-intensity hadron beams.</i> International Particle Accelerator Conference.	2027
<i>High-dimensional maximum-entropy phase space tomography.</i> North American Particle Accelerator Conference. [paper] [slides]	2025

Contributed Talks

<i>Designing experiments to study large space charge tune shifts in the SNS accumulator ring.</i> ICFA Mini-Workshop on Space Charge. [slides]	2026
<i>Four-dimensional phase space tomography from one-dimensional measurements of a hadron beam.</i> International Linear Accelerator Conference. [slides] [code]	2026
<i>Bunch compression experiments at SNS.</i> International Muon Collider Collaboration Annual Meeting. [slides]	2026
<i>Repurposing existing accelerator facilities for Muon Collider R&D.</i> APS Global Physics Summit. [slides]	2026
<i>Proton Driver for the Demonstrator and Muon Collider.</i> US Muon Collider Annual Meeting. [slides]	2025
<i>Bunch rotation experiments and benchmarking at SNS.</i> International Muon Collider Collaboration Meeting. [slides]	2025

<i>SNS accelerator synergies for muon collider R&D.</i> Princeton Muon Collider Organizational Workshop. [slides]	2024
<i>The impact of high-dimensional phase space correlations on the beam dynamics in a linear accelerator.</i> ICFA Workshop on High-Intensity and High-Brightness Hadron Beams. [paper] [slides]	2023
<i>Measurements of the five-dimensional phase space distribution of an intense ion beam.</i> North American Particle Accelerator Conference. [paper] [slides]	2022
<i>Accelerator research and development at the Spallation Neutron Source.</i> Particle Accelerators: Technology and Applications (UTK graduate course). [slides]	2022
<i>Self-consistent, angular-momentum-dominated hadron beams for space charge mitigation.</i> ICFA Mini-Workshop on Space Charge. [slides]	2022
<i>Development of an injection-painted self-consistent beam at the Spallation Neutron Source.</i> ICFA Workshop on High-Intensity and High-Brightness Hadron Beams. [paper]	2021
<i>Computation of the matched envelope of the Danilov distribution.</i> APS April Meeting. [slides]	2021

Posters

<i>Search for Vlasov-Poisson equilibrium distributions using generative models.</i> A. Hoover. International Particle Accelerator Conference. [paper]	2026
<i>Feasibility of proton bunch compression in an operational high-power accelerator.</i> A. Hoover, V. Morozov, C. Piller, A. Narayan. North American Particle Accelerator Conference. [paper]	2025
<i>MENT-Flow: maximum-entropy tomography using normalizing flows.</i> A. Hoover, C. Wong. International Particle Accelerator Conference. [paper]	2024
<i>Detailed characterization of a five-dimensional phase space distribution.</i> A. Hoover, K. Ruisard, A. Aleksandrov, S. Cousineau, A. Zhukov. International Particle Accelerator Conference. [paper]	2023
<i>Simulation of 4D emittance measurement at the Spallation Neutron Source.</i> A. Hoover, N. Evans. International Particle Accelerator Conference. [paper]	2021

Coauthor

<i>Beam loss modeling and mitigation due to intra-beam stripping.</i> S. Kakkar, V. Morozov, L. Lee, N. Evans, A. Zhukov, A. Hoover. North American Particle Accelerator Conference. [paper]	2025
<i>Exploring space charge mitigation with eigenpainting at the SNS.</i> N.J. Evans, A. Hoover, T. Gorlov, V. Morozov. North American Particle Accelerator Conference. [paper]	2025
<i>Progress toward halo modeling at the SNS Beam Test Facility.</i> K. Ruisard, A. Aleksandrov, A. Hoover, T. Thompson, A. Zhukov. Linear Accelerator Conference. [paper]	2024
<i>Proposed muon collider R&D at the SNS.</i> V. Morozov, A. Hoover, D. Stratakis, F. Pilat, N. Evans, S. Cousineau. International Particle Accelerator Conference. [paper]	2024
<i>Measurement and modeling of beam transport in the FODO line of the Spallation Neutron Source Beam Test Facility.</i> T. Thompson, K. Ruisard, A. Hoover, A. Aleksandrov, A. Zhukov. International Particle Accelerator Conference. [paper]	2024
<i>Status of Spallation Neutron Source Beam Test Facility and progress of beam dynamics studies.</i> K. Ruisard, A. Aleksandrov, S. Cousineau, R. Dickson, B. Han, A. Hoover, A. Shishlo, T. Thompson, V. Tzoganis, R. Welton, A. Zhukov. International Particle Accelerator Conference. [paper]	2024
<i>Self-consistent injection painting for space charge mitigation.</i> N. Evans, A. Hoover, T. Gorlov, V. Morozov. ICFA Workshop on High-Intensity and High-Brightness Hadron Beams. [paper]	2023
<i>Measurements at peak operational beam current in the SNS beam test facility.</i> K. Ruisard, A. Hoover, A. Aleksandrov, A. Zhukov, S. Cousineau. International Particle Accelerator Conference. [paper]	2023

<i>Phase space painting of a self-consistent Danilov distribution in the SNS Ring.</i> N. Evans, A. Hoover, T. Gorlov, V. Morozov. International Particle Accelerator Conference. [paper]	2023
<i>Model/measurement comparison of the transverse phase space distribution of an RFQ-generated bunch at the SNS BTF.</i> K. Ruisard, A. Aleksandrov, S. Cousineau, A. Hoover, A. Zhukov. North American Particle Accelerator Conference. [paper]	2022
<i>Observation of current-driven features of 2.5 MeV ion bunch with complete and efficient 5D measurements at the SNS Beam Test Facility.</i> K. Ruisard, A. Aleksandrov, S. Cousineau, A. Hoover, A. Zhukov. International Linear Accelerator Conference. [paper]	2022
<i>Open XAL Status Report 2022.</i> A. Zhukov, A. Hoover, A. Shishlo, J. F. Esteban Müller, E. Laface, Y. Levinsen, N. Milas. International Particle Accelerator Conference. [paper]	2021

Miscellaneous

<i>Six Dimensional Distributions at the SNS Beam Test Facility.</i> K. Ruisard, A. Aleksandrov, A. Hoover. APS Division of Physics of Beams Annual Newsletter. [link]	2022
<i>Towards the production of a self-consistent phase space distribution.</i> A. Hoover. PhD Diss., University of Tennessee. [link]	2022

Seminars

<i>Accelerator and beam physics research at ORNL.</i> Adlai E. Stevenson High School Physics Club.	2026
<i>Accelerator research and development at the Spallation Neutron Source.</i> University of Tennessee Nuclear Engineering Department, graduate course (Particle Accelerators: Technology and Applications).	2022

Teaching Experience

Instructor, United States Particle Accelerator School, <i>Beam Physics with Intense Space Charge</i> . [website]	2026
Teaching Assistant, United States Particle Accelerator School, <i>Accelerator Fundamentals</i> .	2021
Graduate Teaching Assistant, Department of Physics, University of Tennessee.	2018–2019
Teaching Assistant, Department of Physics, Wheaton College.	2018–2019

Service

Program committee, United States Muon Collider Collaboration (USMCC) Annual Meeting	2026
Organizing committee, International Muon Collider Collaboration (IMCC) Annual Meeting	2026
United States Muon Collider Collaboration (USMCC) Leadership Strategy Group (Proton Driver)	2025–
Reviewer for Physical Review Letters (1), Physical Review Accelerators and Beams (2), and Nature Communications Physics (1).	